

Lab Report

Problem Statement

Modern educational platforms contain a large number of tutor profiles with different subjects, teaching styles, prices, and educational levels. Finding the most relevant tutor for a student can therefore become difficult using only traditional keyword search or filtering systems.

The goal of this project is to build a hybrid recommender system capable of recommending tutors based on both:

- semantic similarity between student queries and tutor profiles,
- and collaborative recommendation techniques.

To solve this problem, the project combines:

- Word Embeddings,
- Content-Based Filtering,
- Collaborative Filtering,
- and Hybrid Ranking strategies.

The system aims to provide more personalized and semantically relevant tutor recommendations.

Dataset

The datasets used in this project are custom tutor and ratings datasets created for educational and research purposes.

The dataset `tutor.py` was generated using LLM ChatGPT.

It contains:

- tutor profiles (name,email,location),
- tutor biographies,
- teaching subjects,
- prices,
- educational levels,
- special needs covered

The dataset `ratings.py` was also generated using ChatGPT.

It contains:

- Students ID
- Tutor ID
- ratings given by the student to the tutor

Ratings were simulated as a Gaussian distribution centered around 3.5 and with an $\text{std} = 1$. This type of distribution is common in recommender systems datasets because users tend to avoid giving extremely low scores.

For all your persona when you run the script (`main.py`) you will be the user 10 with 5 previous classes

Source Code

The source code is written in Python and organized into different modules corresponding to the recommendation pipeline.

Main components include:

- preprocessing (cleaning and tokenization),
- embedding generation,
- KNN candidate retrieval,
- recommender systems

Main technologies and libraries:

- Python
- NumPy
- Pandas
- Scikit-learn
- Gensim
- GloVe (unsupervised learning algorithm for obtaining vector representations for words)

Flow

The data of your persona is separated between :

- Personal information (Of no use in this project)
- Ranking features (such as Prices, Subject, Level,...)
- Student's Query

1) Preprocess of the query (preprocess.py) :

- Clean the query
- Tokenize it
- Remove stop words

Output : List of Cleaned Tokens

2) Query embedding. There is 2 model for embedding :

- A made-by-hand word2vec which is trained on the bio of tutors.csv
- A GloVe model trained of wikipedia : <https://nlp.stanford.edu/projects/glove/>
if you want to use the GloVe model you need to download the pre-trained word vectors
([glove.2024.wikigiga.100d.zip](https://nlp.stanford.edu/projects/glove/)) 560 MB download.

When the embedding of each word is finished we create a new vectors equals to the mean of each words of the query (mean pooling)

3) Calculate the similarity between the student query and all the tutor bio.

We use cosine similarity to calculate the semantic closeness of the 2 texts.

Here's they are different way of doing things.

→ You can use KNN of the results to get the K-best tutors (let's say K=100). It is necessary when you start to have a lot of data and you need to speed up the process (commented line 82 in recommender.py)

→ In my case, I didn't use KNN here because I have only 200 Tutors.

Output : Similarities score

4) Integrate other ranking features with the similarities score to get a final score.

→ We do a weighted sum of each score to get the final score.

The weights are manually setup in main.py L.108

```
weights_cb = {
    "similarity": 0.5,    # bio
    "subject": 0.35,
    "level": 0.025,
    "hourly_rate": 0.05,
    "special_needs": 0.025,
    "preferred_learning_mode": 0.05
}
```

Output : ContentBasedFiltering Scores

5) Get the CollaborativeFiltering Scores

- retrieve user feature from the user_id and calculate the raw prediction (thanks to SVD)
- Normalize
- In case of cold start (no ratings), the scores is 0

6) Merge the 2 scores in the Hybrid Model to obtain the final ranking

Output Explanation

Here we are using the pre-generated persona (available when you write 'load')

Important features : The User is looking for a Computer Science teachers specialized in Neural Networks

Made-by-hand Word2Vec :

```
.venv ~/D/H/I/IS_recommenderSystemWithWordEmbeddings → ./main.py
Do you want to create your persona or to load one ? [create/load] : load

--- Summary of your request ---
fullname: Noa Rispal
email: None
location: None
query: I want to learn optimizer for Neural Networks such as Adam Optimizer
subject: Computer Science
price: 60
needs:
mode: Online
level: undergraduate

-----

Data collected successfully !

Now here's the model prediction :

--- Testing Recommendation for User : Noa Rispal ---
Query: 'I want to learn optimizer for Neural Networks such as Adam Optimizer'

Results [Hybrid CB 90.0% / CF 10.0%]:
- Wendy Pham
  → Domain: Computer Science
  → Precisely: Artificial Intelligence
  → Bio: Student working on clustering algorithms, k-means, dimensionality reduction
  → Price: 33
  → Cover needs: ADHD, Dyslexia
- Linh Do
  → Domain: Computer Science
  → Precisely: Data Engineering
  → Bio: Specialist in Kafka streaming, Hadoop ecosystem, data lake architecture for analytics
  → Price: 52
  → Cover needs: Autism
- Duy Pham
  → Domain: Computer Science
  → Precisely: Data Engineering
  → Bio: Backend data specialist handling ETL workflows, Redshift, schema design
  → Price: 48
  → Cover needs: nan
- Guillaume Blanc
  → Domain: Computer Science
  → Precisely: Cybersecurity
  → Bio: Expert in penetration testing, intrusion detection, cryptography enhancing system security
  → Price: 82
  → Cover needs: Autism
- Xuan Dang
  → Domain: Computer Science
  → Precisely: Data Engineering
  → Bio: Engineer working with distributed systems, Spark SQL, data partitioning techniques
  → Price: 51
  → Cover needs: ADHD, Dyslexia

Retry with more serendipity ? [y/n] : █
```

The first results are generated using a low-serendipity model, meaning that the final scores mainly come from the Content-Based Filtering component. We can observe that all the recommended profiles are Computer Science teachers specializing either in Data Engineering or Artificial Intelligence. Since the `Precise_domain` feature is not directly used inside the model (it was only added for demonstration purposes), this highlights the fact that the system extracted these profiles solely based on the semantic similarity between the tutor biographies and the user query. The last teacher is a Cybersecurity teacher who still matches the user's level, subject, and teaching mode (Online). Even though this tutor has no direct connection to Artificial Intelligence, the biography remains strongly related to Computer Science. If we increase serendipity by pressing 'y', we obtain :

```

--- Testing Recommendation for User : Noa Rispal ---
Query: 'I want to learn optimizer for Neural Networks such as Adam Optimizer'

Results [Hybrid CB 30.0% / CF 70.0%]:
- Xuan Dang
  -> Domain: Computer Science
  -> Precisely: Data Engineering
  -> Bio: Engineer working with distributed systems, Spark SQL, data partitioning techniques
  -> Price: 51
  -> Cover needs: ADHD, Dyslexia
- Victor Hugo II
  -> Domain: Computer Science
  -> Precisely: Cybersecurity
  -> Bio: Specialist in penetration testing, encryption, incident response strengthening enterprise security
  -> Price: 85
  -> Cover needs: Autism
- Jack Vu
  -> Domain: Physics
  -> Precisely: Quantum Physics
  -> Bio: Master's student studying quantum field theory, Dirac equation, particle spin interactions
  -> Price: 38
  -> Cover needs: nan
- Guillaume Blanc
  -> Domain: Computer Science
  -> Precisely: Cybersecurity
  -> Bio: Expert in penetration testing, intrusion detection, cryptography enhancing system security
  -> Price: 82
  -> Cover needs: Autism
- Wendy Pham
  -> Domain: Computer Science
  -> Precisely: Artificial Intelligence
  -> Bio: Student working on clustering algorithms, k-means, dimensionality reduction
  -> Price: 33
  -> Cover needs: ADHD, Dyslexia

```

Here we can observe two new tutors appearing in the recommendations: Victor Hugo II and Jack Vu. Even if Victor Hugo II was probably already very close to the first results (perhaps within the top 10) the fact that he now appears in the top recommendations suggests that users with tastes similar to the current user either positively interacted with his profile or negatively interacted with a previously recommended top-5 tutor.

However, the most interesting recommendation is Jack Vu, who is a Physics teacher and therefore appears almost exclusively because of the collaborative component of the system and the social relationships learned from user interactions.

GloVe :

```

--- Testing Recommendation for User : Noa Rispal ---
Query: 'I want to learn optimizer for Neural Networks such as Adam Optimizer'

Results [Hybrid CB 90.0% / CF 10.0%]:
- Alice Nguyen
  -> Domain: Computer Science
  -> Precisely: Artificial Intelligence
  -> Bio: Researcher working on CNN, backpropagation, Adam optimizer with focus on computer vision systems
  -> Price: 45
  -> Cover needs: ADHD, Dyslexia
- Elise Morel
  -> Domain: Computer Science
  -> Precisely: Artificial Intelligence
  -> Bio: Doctoral candidate studying deep learning, dropout regularization, batch normalization
  -> Price: 52
  -> Cover needs: ADHD, Dyslexia
- Yasmine Benali
  -> Domain: Computer Science
  -> Precisely: Artificial Intelligence
  -> Bio: AI enthusiast experimenting with LSTM networks, sequence modeling, attention mechanisms
  -> Price: 36
  -> Cover needs: nan
- Giang Pham
  -> Domain: Computer Science
  -> Precisely: Artificial Intelligence
  -> Bio: Graduate student applying reinforcement learning, Q-learning, policy gradients to robotics problems
  -> Price: 35
  -> Cover needs: Dyslexia
- Laura Petit
  -> Domain: Computer Science
  -> Precisely: Web development
  -> Bio: UI developer skilled in HTML5, Sass, responsive design principles
  -> Price: 38
  -> Cover needs: ADHD, Dyslexia

Retry with more serendipity ? [y/n] :

```

The results we obtained with the GloVe embedding are better. Since GloVe is trained on wikipedia it has a lot more good quality data to train on and thus is more precise. However, because this corpus is not specialized for educational recommendation tasks, the semantic space remains relatively broad. As a consequence, the embedding of a Web development teacher can still be relatively close to that of an AI teacher, simply because both domains belong to the larger Computer Science field and must remain semantically distant from completely unrelated topics such as “Dog Training” pages on Wikipedia.

Conclusion

These results demonstrate that the hybrid model is functioning correctly, since it is capable of introducing new profiles that are not directly linked to the initial semantic query. Nevertheless, the system still has several limitations that would require larger and more realistic datasets, as well as the training of a domain-specific embedding model, in order to improve recommendation quality.

Moreover, serendipity is not necessarily desirable during the user’s first interactions with the platform, since the student is generally searching for a very precise topic. However, as the number of interactions increases, the system should progressively increase serendipity in order to help the user discover new subjects, tutors, and learning opportunities outside of their initial search space.